III B.Tech – II Semester

Course Code	**BUILDING INFORMATION MODELING**	L	T	P	C
23A01607		0	1	2	2

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamentals of Building Information Modeling (BIM) and Autodesk Revit.
2. **Develop** proficiency in Revit's basic drawing and editing tools for structural and architectural modeling.
3. **Create** 3D models of buildings, including walls, floors, ceilings, roofs, stairs, and railings.
4. **Analyze** different components such as curtain walls, doors, windows, and structural elements.
5. **Apply** various visualization and detailing techniques to generate callouts, elevations, and sections.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Explain** the fundamentals of BIM and Autodesk Revit's interface and workflow.
2. **Use** basic drawing, editing, and modificationtoolsin Revit for creating and modifying models.
3. **Model** various architectural elements such as walls, doors, windows, floors, ceilings, and roofs.
4. **Construct** structural elements including grids, columns, stairs, railings, and ramps.
5. **Generate** 3D views, sections, and elevations for visualization and detailing purposes.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	3	1	-	-	-	1	-	2	3	3
CO -2	3	3	3	3	3	2	-	-	-	-	-	2	3	3
CO -3	3	3	3	3	3	3	1	-	-	-	-	2	3	3
CO -4	3	3	3	3	3	3	1	-	-	-	-	2	3	3
CO -5	2	2	2	2	3	3	1	-	-	1	1	2	2	2

List of Experiments: -

1. INTRODUCTION to BIM & AUTODESK REVIT - About AutodeskandAutocad, Workflow and BIM, Revit Terms, Overview of The Interface, Starting Projects, Viewing Commands.
2. BASIC DRAWING and EDITINGtoOLS - Using General Drawing tools, Editing Elements, Working With Modification tools.
3. SETTING UP LEVELS and GRIDS - Setting up Levels and Grids, Creating Structural Grids, Adding Columns, Linking and Importing CAD files.
4. MODELING WALLS ModellingWalls, Modifying Walls, Model Exterior Shell, Add Interior Walls.
5. WORKING WITH DOORS and WINDOWS Inserting Doors and Windows, Loading Door and Window Types From Library, Creating Additional Door and Window Sizes.

6. WORKING WITH CURTAIN WALLS Creating Curtain Walls, Adding Curtain Grids, Working With Curtain Wall Panels, Attaching Mullions to Curtain Grids.
7. WORKING WITH VIEWS Setting The View Display, Duplicating Views, Adding Callout Views, Elevations and Sections.
8. ADDING COMPONENTS Adding Component, Modifying Component, Working With Elements.
9. MODELING FLOORS Modelling& Modifying Floors, Joining Geometry, Creating Shaft Openings, Creating Sloped Floors
10. MODELING CEILINGS & ROOFS ModellingCeilings, Adding Ceiling Fixtures, Creating Ceiling Soffits, Modelling Roofs
11. MODELING STAIRS and RAILING Creating Component Stairs, Modifying Component Stairs, Working With Railings, Sketching Custom Stairs, Creating Ramps.

TEXT BOOKS:

1. Chuck Eastman, Paul Teicholz, Rafael Sacks, Kathleen Liston ―BIM HANDBOOK, Wiley, 2nd Edition, 2011
2. Wing, Eric. Autodesk Revit Architecture 2017: No Experience Required. Indianapolis: John Wiley & Sons, 2016

REFRENCE BOOKS:

1. Kim, Marcus, Lance Kirby, and Eddy Krygiel. Mastering Autodesk Revit 2017 for architecture. 1st ed. INpolis, IN: John Wiley & Sons, 2016.
2. Garber, Richard. BIM Design: Realizing the Creative Potential of Building Information Modeling. AD Smart 02. Chichester, U.K.: Wiley, 2004
3. Peter B. and Nigel D., "BIMin Principle and in Practice", 1 st Edition, ICE Publishing, 2014.
4. BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors,
5. Chuck Eastman, Paul Teicholz, Rafael Sacks and Kathleen Liston, John Wiley & Sons, 2008.
6. BIM and Construction Management: Proven tools, Methods, and Workflows, Brad Hardin, Sybex, 2009.
7. Building Information Modeling: BIMin Current and Future Practice, Karen Kensek and Douglas Noble, Wiley, 2014, First Edition.

Online Learning Resources:

https://minnodillc.com/building-information-modeling-bim/
https://www.skyfilabs.com/online-courses/building-information-modelling-course